
Tuesday, 2 June 2026

Energy Market Analytics Portfolio and Capstone Direction

Technical Work, Current Projects, and Future Research Direction

SPEAKER

Vivek Bharadwaj Aryasomayajula

MEETING CONTEXT

Research Centre Director Progress & Capstone
Supervision Review

PURPOSE OF THIS DISCUSSION

Framework for progress alignment, learning progression, and capstone direction approval.

Presentation Overview

- Technical work completed over recent months
- Current energy market analytics projects
- Strategic alignment into a broader research pathway
- Proposed capstone direction
- Future opportunities in analytics and modelling

Key Objective

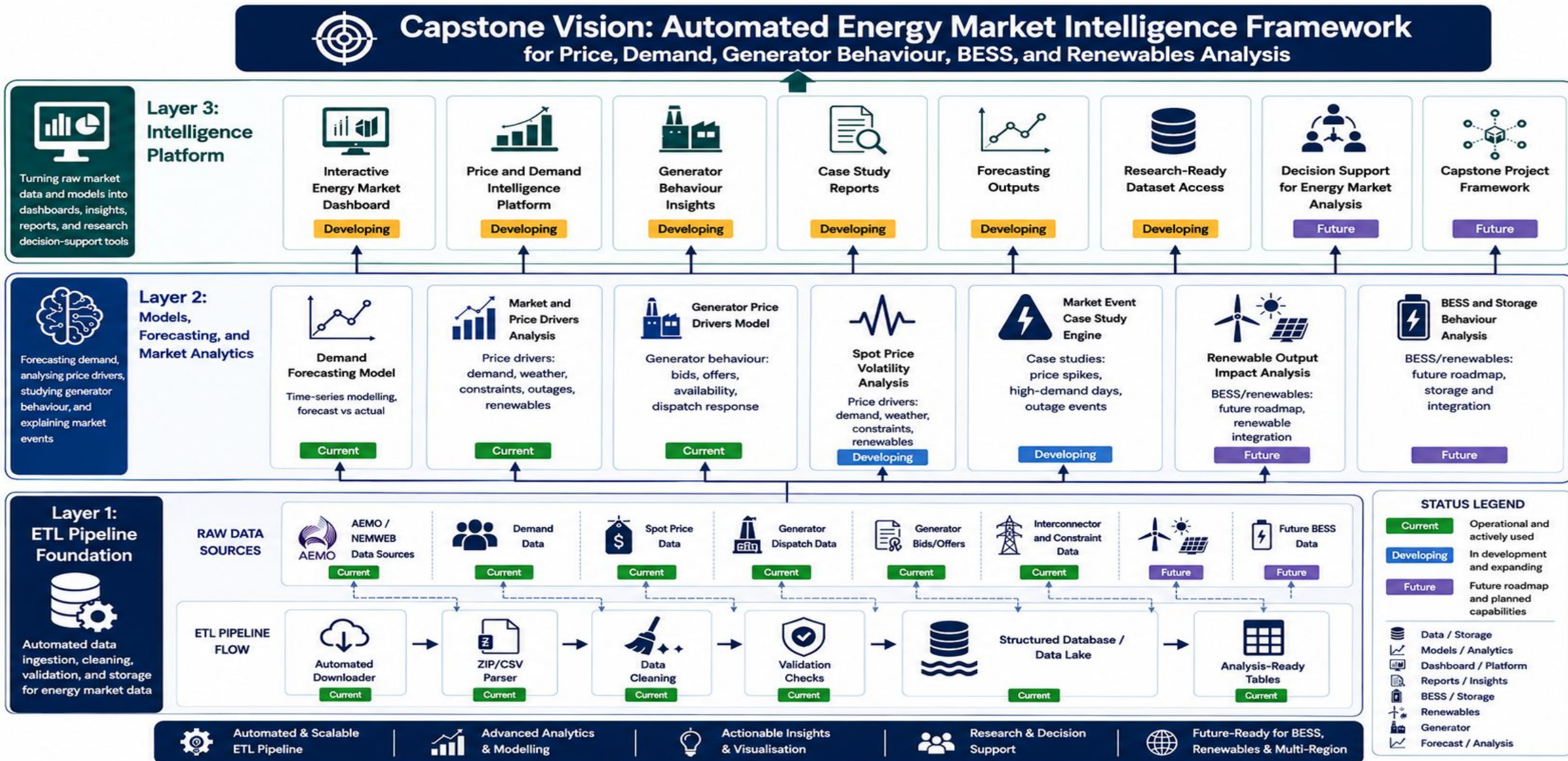
Evolve work into a stronger research contribution and practical energy market intelligence capability.

Research Roadmap

Past Work → Current Projects → Capstone → Future Research

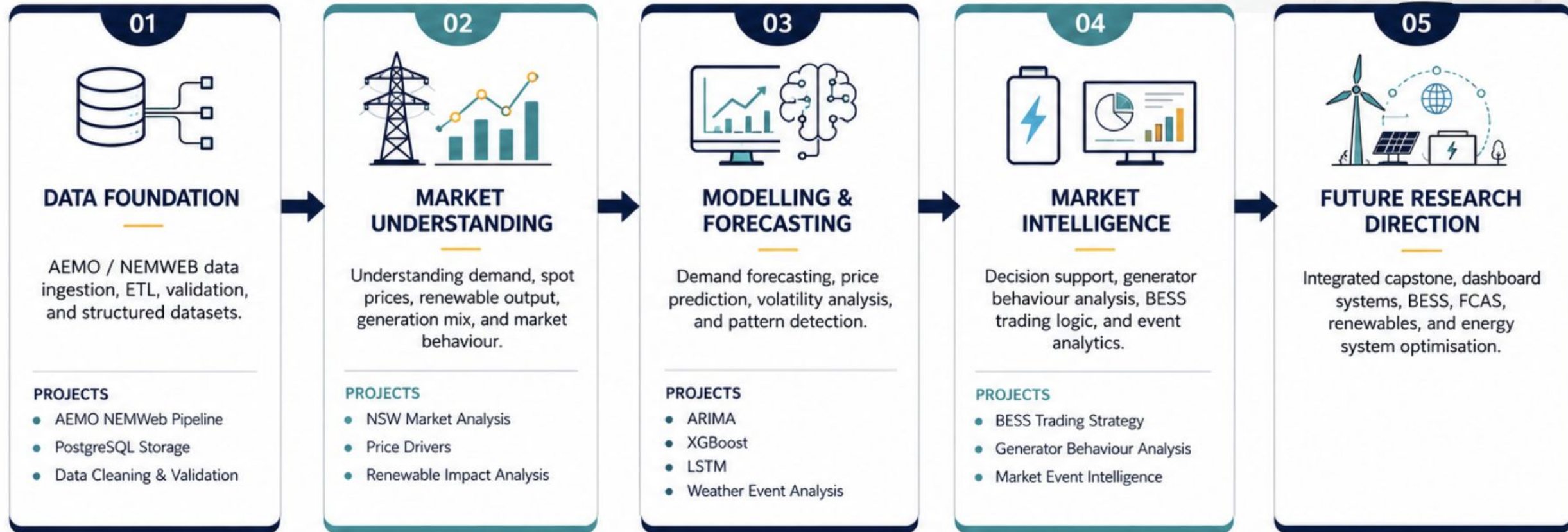
BIG PICTURE PROJECT ECOSYSTEM

Connected Energy Market Analytics Ecosystem — Core Research Journey



ENERGY MARKET ANALYTICS METHODOLOGY

A structured analytical framework used across energy market projects



This methodology transforms raw AEMO market data into market understanding, forecasting capability, and future decision intelligence.

STRATEGIC PROJECT OVERVIEW (IN PROGRESS / PLANNED)

Goal: Build a connected, research-oriented NSW Energy Market Analytics ecosystem rather than isolated projects.

Initiative	Overview	Strategic Focus	Status
Integrated NSW Energy Market Intelligence Platform	End-to-end analytics ecosystem connecting forecasting, market behaviour, price drivers, and decision intelligence	Energy Market Intelligence	 In Progress
Automated AEMO/NEMWeb Data Pipeline Framework	Research-ready ETL framework for automated energy market data extraction and validation	Data Engineering & Research Infrastructure	 In Progress
NSW Price & Demand Intelligence Capstone	Integrated analytical framework for combining demand forecasting, price drivers, generator behaviour, and case studies	Capstone Research	 Planned
Market Decision Intelligence System	Analytics layer to interpret price spikes, volatility, constraints price drivers, generator behaviour, and case studies	Market Intelligence	 In Progress
Future Energy Systems Analytics (BESS & Renewables)	Future roadmap exploring storage dispatch, FCAS, arbitrage, and renewable events.	Future Energy Systems	 Planned

KEY MESSAGE

Rather than isolated projects, this work is being developed as a **connected energy market analytics ecosystem**.

CONTRIBUTING TECHNICAL PROJECTS (BUILDING BLOCKS)

Goal: Build a connected, research-oriented NSW Energy Market Analytics ecosystem rather than isolated projects.

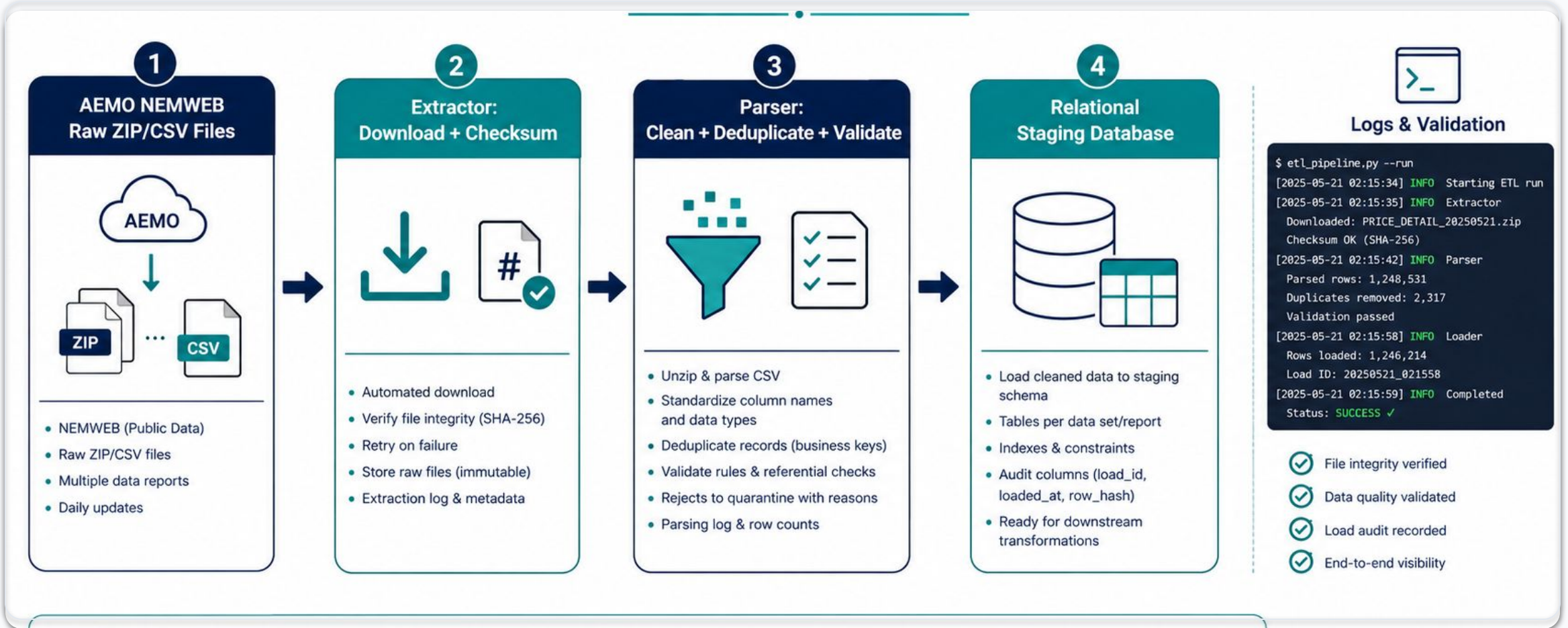
Project	Contribution	Area	Progress
Market Fundamentals & Price Drivers	Understanding demand, PV, supply imbalance, and RRP formation	Price Behaviour	 Completed
Electricity Demand Forecasting	ARIMA → XGBoost → LSTM forecasting workflow	Forecasting	 Completed
BESS Trading Strategy (D-1)	Charge/discharge optimisation logic using market prices	Trading Analytics	 Completed
Renewable Impact Analysis	Solar and wind variability impact on demand and pricing	Grid Behaviour	 In Progress
Volatility & Risk Analysis	Detecting price spikes and abnormal market events	Risk Analytics	 In Progress
AEMO NEMWeb Data Pipeline	Automated extraction, parsing, validation, PostgreSQL loading	Data Engineering	 Completed

KEY MESSAGE

Rather than isolated projects, this work is being developed as a **connected energy market analytics ecosystem**.

ETL PIPELINE ARCHITECTURE

System architecture diagram showing standard AEMO data transformations and staging layers.



Data Integrity

Checksums and validation at every stage.

Reliability

Automated retries, logging and monitoring.

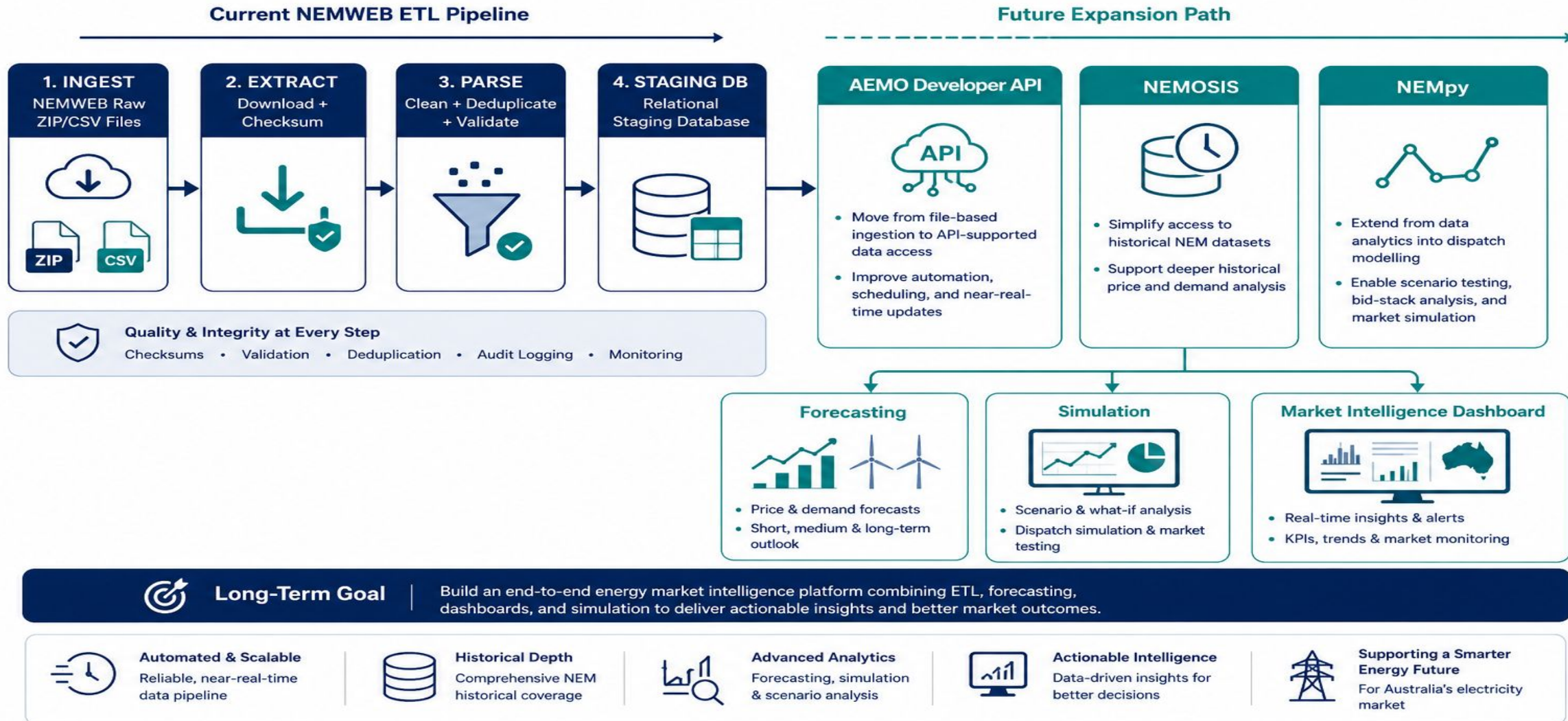
Traceability

Audit logs, level-lineage tracking.

Australian Context

Built for AEMO NEM data and market needs.

Future Expansion



FUTURE RESEARCH ROADMAP

Phased execution timeline detailing immediate delivery and future research extensions.

Short-Term

Q2 2026

Validate and finalise forecasting models.

Ingest Q1 2026 price data to update validation indices.

Medium-Term

Q3 2026

Integrate Streamlit dashboard.

Package forecasting and bid stack tools for internal testing.

Capstone Complete

Q4 2026

Submit and publish findings on NSW wholesale electricity market drivers.

Future Extensions

Model BESS dispatch arbitrage, FCAS co-optimisation, and virtual power plant dispatch paths.

Long-term research alignment: Building this end-to-end framework lays a strong foundation for future research, positioning our center to lead on critical topics like battery optimization, virtual power plants (VPPs), and FCAS market dynamics.

| THANK YOU

Questions & Feedback

Vivek Bharadwaj Aryasomayajula

Energy Market Analytics Portfolio and Capstone Direction

